

smartRoute:

Transforming Urban Commutes with AI-Enabled Multimodal Transit

ABSTRACT PRESENTATION

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Introduction

- Rapid urbanisation has diversified urban transportation options
- Buses and railway networks lack interoperability, fragmenting the commuter experience
- Passengers face challenges in journey planning, route transfers, and accessing real-time information
- The goal is to develop an intelligent mobility application to consolidate all transit networks scheduling into a single interface
- Maps and live schedules are combined into one easy-to-use app.
- Delivers optimised journey planning to users while providing cities with data-driven transit insights

Problem Statement

Independent buses, metro services and transportations force travelers to use multiple applications, creating a confusing and inefficient journey.

Description of Project

- Develop an AI-enabled MaaS platform integrating multiple transport modes.
- Use GTFS(general transit feed specification), GIS(geographic information system), and AI for seamless integration.
- Provide multimodal journey planning and route visualization.
- Offer fare estimation and recommendations.
- Include a mobility analytics dashboard for travel and network insights.
- Support better travel planning and transport decision-making.

Scope of the Project

- Enable multimodal journey planning, route optimization, fare estimation, and transfer guidance through an AI-powered travel assistant
- The system will also include mobility analytics for informed decision-making, with thorough validation of its accuracy, usability, and performance.

Methodologies

- Use data conversion tools to turn raw transit schedules into standard General Transit Feed Specification (GTFS) files.
- Use PostgreSQL with PostGIS extensions to store, index, and quickly look up map shapes and coordinates.
- Use Dijkstra's Algorithm to find the shortest path through a network of stations.

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- Use Round Robin scheduling (RR) to handle incoming search requests .
- Build a Progressive Web App (PWA) that loads instantly and works offline.
- Test and validate the system using real-world scenarios and performance metrics.

System Features

- Multimodal journey planning across different transport modes
- GTFS-based transport data integration
- GIS-based route and network visualization
- AI-powered travel assistant for queries and recommendations
- Fare estimation and transfer guidance
- Personalized route suggestions
- Mobility analytics dashboard for travel patterns and performance
- User-friendly PWA interface for easy access and navigation

Software Requirements

- Operating System: Windows 11
- Frontend: React
- Backend: Python
- Database: PostgreSQL with PostGIS
- Data Format: GTFS
- AI/ML Tools: Python, Pandas
- Routing Engine: OpenTripPlanner

Hardware Requirements

- Processor: Intel i5
 - RAM: Minimum 8 GB
 - Storage: At least 256 GB SSD
 - GPU: Optional (for AI processing, if required)
 - Network: Stable internet connection for real-time data access
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Conclusion

The proposed AI-enabled MaaS platform will integrate multiple transportation modes into a single system, enabling seamless journey planning, route optimization, and travel assistance for urban commuters.

By combining GTFS, GIS, AI, and mobility analytics, the platform supports efficient, user-friendly, and sustainable urban transportation while providing valuable insights for transportation planning and decision-making.

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Thank You
